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Applications Note

Sequence analysis

MethRaFo: MeDIP-seq methylation estimate using a Random Forest Regressor

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Abstract

Motivation: Profiling of genome wide DNA methylation is now routinely performed when studying development, cancer and several other biological processes. Although Whole genome Bisulfite Sequencing provides high-quality methylation measurements at the resolution of nucleotides, it is relatively costly and so several studies have used alternative methods for such profiling. One of the most widely used low cost alternatives is MeDIP-Seq. However, MeDIP-Seq is biased for CpG enriched regions and thus its results need to be corrected in order to determine accurate methylation levels.

Results: Here we present a method for correcting MeDIP-Seq results based on Random Forest regression. Applying the method to real data from several different tissues (brain, cortex, penis) we show that it achieves almost 4 fold decrease in run time while increasing accuracy by as much as 20% over prior methods developed for this task.

Availability and implementation: MethRaFo is freely available as a python package (with a R wrapper) at https://github.com/phoenixding/methrafo.

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Supplementary information: Supplementary data are available at *Bioinformatics* online.

1 Introduction

DNA Methylation is a process by which methyl groups are added to the DNA molecule. While not all functional aspects of methylation are currently known, it is already clear that it plays an important role in regulating gene expression and protein–DNA interactions. Abnormal DNA methylation has been recognized as an important factor in diseases including diabetes (Slomko *et al.*, 2012) and cancer (Ehrlich *et al.*, 2002) and methylation was also shown to impact aging (Richardson *et al.*, 2003) and other biological processes. These findings have led to wide interest in genome wide methylation profiling which is now routinely used when studying biological processes and diseases.

These studies rely on methods that have been recently developed for such genome wide methylation profiling including whole-genome bisulfite sequencing (BS-Seq) (Krueger *et al.*, 2012) and enrichment-based methods such as MeDIP-Seq and MRE-Seq (Stevens *et al.*, 2013). Methylation profiling results from the BS-Seq are considered the most accurate providing resolution at the single base level (Riebler *et al.*, 2014). For BS-Seq single-base readouts preserve methylated cytosines while unmethylated cytosines are all converted to uracil (Susan *et al.*, 1994). However, BS-Seq is more than 3 times more expensive when compared with enrichment-based methods such as MeDIP-Seq (Jeong *et al.*, 2016). Another advantage of MeDIP-Seq is that unlike bisulfite sequencing based methods it can distinguish between DNA methylation and 5-hydroxymethylation (Li *et al.*, 2015). Thus, although MeDIP-Seq offers a lower resolution and is biased towards hypermethylated regions, it is still widely used in studies that profile a large number of samples, for example time series studies or studies comparing different patient samples, because of its ability to profile a large number of samples at a fraction of the cost of BS-Seq (Yong *et al.*, 2016). Given the wide use, correcting for MeDIP-Seq biases is an important and timely problem.

To address this need, a number of MeDIP-Seq methylation correction methods have been developed in the past few years.

BATMAN (Down *et al.*, 2008) addresses the problems of the bias MeDIP-Seq has for dense CpG regions by using a linear function to normalize its results based on the local CpG density. The other two widely used MeDIP-Seq methylation correction tools, MEDIPS (Lienhard *et al.*, 2014) and BayMeth (Riebler *et al.*, 2014) also attempt to correct for local CpG density and differ in the specific type of functions used for such correction (see Supplementary Material). Although these methods improve the correlation to the gold standard BS-Seq result, their run time and memory usage are extensive making it hard to use them to correct a large batch of samples. For example, we tested BATMAN, MeDIPS and BayMeth on a single genome wide breast luminal epithelial cell MeDIP-Seq dataset from the roadmap epigenomics (Bernstein *et al.*, 2010) database. BATMAN was unable to provide the full corrected dataset even after running for 1 week (we used an Intel(R) Core(TM) i7 CPU870 @ 2.93GHz and 16G RAM). MEDIPS and BayMeth took roughly an hour (54 min and 94 min for MEDIPS and BayMeth respectively) and used about 12G RAM. Thus, reducing the run time and memory usage while not decreasing (or even increasing) accuracy is an important goal given the large number of current and future genome wide methylation studies. In this work, we propose a method for such correction that is based on a random forest regressor trained with datasets that were profiled using both, BS-Seq and MeDIP-Seq.

2 Materials and methods

The MethRaFo package provides functions to estimate genome-wide methylation based on MeDIP-Seq data. It can be used with two different types of inputs: Aligned reads (typically bam files) or RPKM (typically wiggle files). Figure 1 presents the flowchart of MethRaFo. When using bam files as the input, MethRaFo first calculates the RPKM, which represents the normalized reads count (against all mapped reads) at each nucleotide. Next, following (Down *et al.*, 2008) we calculate the local CpG density for a given window size. Finally, we train a Random Forest regressor by bootstrapping the input data using the nucleotide level RPKM and the local CpG density (for different resolutions centered at a nucleotide) as potential features for each tree. Note that while we only have two types of features (CpG density and RPKM levels) given the fact that we use different resolutions, and the fact that these are continuous values, the actual number of features is much larger. For training we use datasets that were profiled by both, MeDIP-Seq and BS-Seq methylation data with the latter representing the ground truth. Using the trained model, we can predict the corrected methylation using the calculated RPKM and local CpG density at each nucleotide. The MethRaFo package provides 4 functions: methrafo.download, methrafo.bamScript, methrafo.train and methrafo.predict. methrafo.download is used to download the reference genome (e.g. hg19,mm10) needed for further analysis. methrafo.bamScript is used to process bam files. It can be used to calculate genome-wide RPKM (in bigWig format) for any given bam files. methrafo.train is used to train the random forest regressor using calculated features. methrafo.predict is used to predict methylation based on calculated features and the trained model.

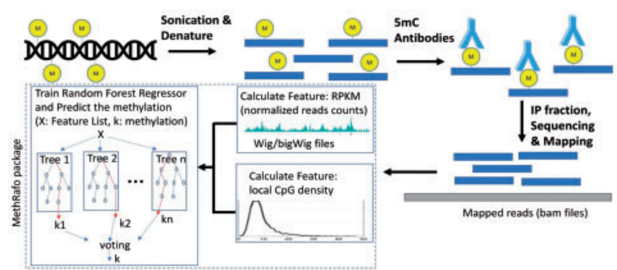

Fig. 1. MethRaFo flowchart

3 Application

To demonstrate the functionality and performance of the MethRaFo package, we trained our model on the Breast_Luminal_Epithelial_Cells dataset from the Roadmap epigenetic datasets (Bernstein *et al.*, 2010) and then used the model learned from this data to correct genome wide methylation values obtained by MeDIP-Seq for three other datasets: Brain Germinal Matrix (brain), Neurosphere Cultured Cells Cortex Derived (cortex) and Penis Foreskin Keratinocyte Primary Cells (penis). In all cases we compared the results to the BS-Seq results for these cells. Table 1 presents the results of this analysis and compares the results of MethRaFo to results from prior methods (as noted above, we were unable to compare to BATMAN given its long runtime). As can be seen, MethRaFo is much faster and more efficient (in terms of memory usage) than the other two methods for all three datasets. This does not come at the expense of accuracy. In fact, for two of the three datasets MethRaFo outperforms that other two datasets by an average of more than 10% for the Spearman correlation and more than 20% for the Pearson correlation between the corrected and BS-Seq results. For the third datasets (brain) it achieves comparable results in terms of Spearman correlations and better results in terms of Pearson correlations. To conclude, MethRaFo presents a practical, efficient and accurate solution for correcting MeDIP-Seq methylation data. MethRaFo was only trained and tested for human data due to lack of appropriate mouse training data. While we believe that the method can be equally applied to mouse, we would recommend training it on mouse data first, if it becomes available, rather than directly using the human trained version.

Table 1. MethRaFo performance comparison

Dataset	Measurement	MethRaFo	MEDIPS	BayMeth	Original
Cortex	Spearman correlation	0.426	0.375	0.318	0.378
Cortex	Pearson correlation	0.717	0.572	0.418	0.325
Cortex	Running time & RAM	16m30s(4G)	54m29s(12G)	97m12s(12G)	–
Penis	Spearman correlation	0.465	0.378	0.320	0.384
Penis	Pearson correlation	0.671	0.463	0.370	0.261
Penis	running time & RAM	14m33s(4G)	60m30s(12G)	89m27s(12G)	–
Brain	Spearman correlation	0.416	0.420	0.386	0.408
Brain	Pearson correlation	0.544	0.501	0.491	0.467
Brain	Running time & RAM	15m23s(4G)	56m12s(12G)	93m23s(12G)	–

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Conflict of Interest: none declared.

References

Bernstein,B.E. *et al*. (2010) The NIH roadmap epigenomics mapping consortium. *Nat. Biotechnol*., **28**, 1045–1048.

Down,T.A. *et al*. (2008) A Bayesian deconvolution strategy for immunoprecipitation-based DNA methylome analysis. *Nat. Biotechnol*., **26**, 779–785.

Ehrlich,M. (2002) DNA methylation in cancer: too much, but also too little. *Oncogene*, **21**, 5400.

Issa,J.P. *et al*. (1996) Switch from monoallelic to biallelic human IGF2 promoter methylation during aging and carcinogenesis. *Proc. Natl. Acad. Sci. USA*, **93**, 11757–11762.

Jeong,H.M. *et al*. (2016) Efficiency of methylated DNA immunoprecipitation bisulphite sequencing for whole-genome DNA methylation analysis. *Epigenomics*, **8**, 1061–1077.

Krueger,F. *et al*. (2012) DNA methylome analysis using short bisulfite sequencing data. *Nat. Methods*, **9**, 145–151.

Li,D. *et al*. (2015) Combining MeDIP-seq and MRE-seq to investigate genome-wide CpG methylation. *Methods*, **72**, 29–40.

Lienhard,M. *et al*. (2014) MEDIPS: genome-wide differential coverage analysis of sequencing data derived from DNA enrichment experiments. *Bioinformatics*, **30**, 284–286.

Richardson,B. *et al*. (2003) Impact of aging on DNA methylation. *Ageing Res. Rev*., **2**, 245–261.

Riebler,A. *et al*. (2014) BayMeth: improved DNA methylation quantification for affinity capture sequencing data using a flexible Bayesian approach. *Genome Biol*., **15**, R35.

Slomko,H. *et al*. (2012) Minireview: epigenetics of obesity and diabetes in humans. *Endocrinology*, **153**, 1025–1030.

Stevens,M. *et al*. (2013) Estimating absolute methylation levels at single-CpG resolution from methylation enrichment and restriction enzyme sequencing methods. *Genome Res*., **23**, 1541–1553.

Susan,J.C. *et al*. (1994) High sensitivity mapping of methylated cytosines. *Nucleic Acids Res*., **22**, 2990–2997.

Yong,W.S. *et al*. (2016) Profiling genome-wide DNA methylation. *Epigenet. Chrom*., **9**, 26.